Yujie Cen

PhD Candidate · Computational Materials Science · TU Wien
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RESEARCH FOCUS

Computational thermal transport and MD simulation in low-dimensional nanomaterials; machine-learning interatomic potentials (MLIP) for large-scale atomistic simulation; phonon dynamics and heat conduction in quasi-1D nanostructures via line group symmetry analysis.

TECHNICAL SKILLS

MLIP Frameworks	MACE · NequIP · Allegro · NEP (GPUMD)
DFT / MD Codes	VASP · GPUMD · LAMMPS
Python Ecosystem	PyTorch · ASE · Phonopy · pymatgen · NumPy · SciPy
Thermal Transport	Green-Kubo · GKMA · HNEMA
HPC & Dev Tools	SLURM · Git · Linux · LaTeX

EDUCATION

PhD in Materials Chemistry · TU Wien, Austria · Jul 2023 – Present *(expected late 2026)*
Supervisor: Prof. Georg K. H. Madsen · Institute of Materials Chemistry (E165), TU Wien
Co-supervisor: Dr. Jesús Carrete Montaña · Instituto de Nanociencia y Materiales de Aragón, CSIC-Universidad de Zaragoza, Spain

MS in Theoretical Physics · South China University of Technology · Sept 2020 – Jul 2023

BS in Physics · South China University of Technology · Sept 2016 – Jul 2020

RESEARCH EXPERIENCE

PhD Candidate · TU Wien · Jul 2023 – Present

Symmetry-adapted MLIP-driven thermal transport in quasi-1D systems *(in preparation)*
I extend Green-Kubo Modal Analysis (GKMA) and HNEMA by replacing the standard eigenvector basis with line-group projector operators, giving every phonon mode a well-defined irrep label. I trained a NEP force field for (10,0) CNT via a two-round active learning scheme (VASP/PBE-sol-D3 labelling, NEP89 as foundation model), achieving energy RMSE 1.04 meV atom^{-1} and force RMSE 120 meV Å^{-1}. Production GPUMD trajectories (10–40 ns, 200–400 K) validate that irrep-summed κ = 2538 W m^{-1} K^{-1} is invariant across basis constructions, demonstrating physically interpretable symmetry-resolved thermal conductivity.

Ab-initio heat transport in defect-laden quasi-1D systems *(npj Computational Materials, 2026, first author)*
Combined DFT phonon calculations, NEP/GPUMD-accelerated MD, and symmetry-adapted Green-Kubo analysis to quantify how point defects activate symmetry-forbidden inter-channel heat transport in CNTs and WS_2/MoS_2 nanotubes. Demonstrated that MLIP-driven MD reproduces DFT-level thermal conductivity while enabling defect-concentration studies at scales inaccessible to ab initio MD.

pulgon_tools *(Computer Physics Communications, submitted, first author)*
Lead developer of an open-source Python package for automated line-group detection, irrep decomposition, character table generation, and symmetry-adapted force constant correction in quasi-1D systems. Production-quality codebase with full test suite, CI/CD, and PyPI release; the equivariant basis construction is directly analogous to the SO(3)-equivariant representations used in MACE and NequIP. github.com/pulgon-project/pulgon_tools

Determining ground states of alloys by symmetry-based classification *(Physical Review Materials, 2022, first author)*
Developed a symmetry-based enumeration scheme using DFT (VASP) to identify ground-state configurations in multi-component alloys, reducing computational cost by orders of magnitude compared to exhaustive search.

Research Assistant · South China University of Technology · 2020 – 2023

Applied first-principles methods (VASP) to study magnetic phase transitions in La–Mn perovskite membranes under strain and doping, and investigated intrinsic defects in topological material $MnSb_2Te_4$.

PUBLICATIONS

First-Author Papers

1. **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. (2026). Ab-initio heat transport in defect-laden quasi-1D systems from a symmetry-adapted perspective. *npj Computational Materials*, 12(1), 19.
2. **Cen, Y.**, Wieser, S., Madsen, G. K. H., & Carrete, J. (2025). Pulgon-tools: A toolkit for analysing and harnessing symmetries in quasi-1D systems. *Computer Physics Communications* (submitted). arXiv:2603.28435.
3. **Cen, Y.**, He, C. C., Qiu, S. B., Zhao, Y. J., & Yang, X. B. (2022). Determining ground states of alloys by a symmetry-based classification. *Physical Review Materials*, 6(5), L050801.

Co-authored Papers

4. Wieser, S., **Cen, Y.**, Madsen, G. K. H., & Carrete, J. (2025). Accelerating first-principles molecular-dynamics thermal conductivity calculations for complex systems. *Journal of Chemical Theory and Computation*.
5. Zhu, C., Yan, X., **Cen, Y.**, He, C. C., Zhao, Y. J., & Yang, X. B. (2025). Accelerating the identification of stable configurations in mixed-anion perovskite materials. *Computational Materials Science*, 248, 113564.
6. Li, D. Z., **Cen, Y.**, Wang, X., & Yang, X. B. (2024). Residual entropy of ice: A study based on transfer matrices. *Physical Review B*, 110(19), 195414.
7. Xiong, J., Peng, Y. H., Lin, J. Y., **Cen, Y.**, Yang, X. B., & Zhao, Y. J. (2023). High Concentration Intrinsic Defects in $MnSb_2Te_4$. *Materials*, 16(15), 5496.
8. Lin, J. Y., Chen, Z. J., Li, G. L., Zeng, J., **Cen, Y.**, et al. (2022). Theoretical study of magnetic phase transition in $La_{2/3}M_{1/3}MnO_3$ (M = Ca, Sr) membranes. *Physics Letters A*, 432, 128010.

Software

Cen, Y., Wieser, S., Madsen, G. K. H., & Carrete, J. pulgon_tools: A Python package for line group symmetry analysis of quasi-1D nanostructures. github.com/pulgon-project/pulgon_tools

SKILLS & LANGUAGES

Programming: Python (advanced) · Bash · C++ (basic)

Languages: Chinese (native) · English (professional — academic writing and presentations)

Open Source: Active contributor to PULGON project; pulgon_tools package maintainer